

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12416439
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Kitagawa; Natsuko et al.

Refrigerant cycle apparatus

Abstract

A refrigerant cycle apparatus includes a refrigerant circuit that circulates a refrigerant, and a leak sensor that detects a refrigerant leaking from the refrigerant circuit, in which the refrigerant cycle apparatus includes, as an operating mode, a recovery mode for recognizing occurrence of an abnormality in the leak sensor and recovering a refrigerant to a predetermined location in the refrigerant circuit.

Inventors: Kitagawa; Natsuko (Osaka, JP), Toyota; Ryuuichi (Osaka, JP), Fukagawa; Kazunari (Osaka, JP), Nagahara; Keiya (Osaka, JP), Kawano; Satoshi (Osaka, JP), Inoue; Makoto (Osaka, JP), Cornelis; Kevin (Ostend, BE)

Applicant: DAIKIN INDUSTRIES, LTD. (Osaka, JP); DAIKIN EUROPE N.V. (Ostend, BE)

Family ID: 1000008819838

Assignee: DAIKIN INDUSTRIES, LTD. (Osaka, JP)

Appl. No.: 17/973701

Filed: October 26, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230052745 A1	Feb. 16, 2023

Related U.S. Application Data

continuation parent-doc WO PCT/JP2020/019945 20200520 PENDING child-doc US 17973701

Publication Classification

Int. Cl.: F25B49/02 (20060101)

U.S. Cl.:

CPC **F25B49/022** (20130101); F25B2500/222 (20130101); F25B2500/27 (20130101);
F25B2600/2515 (20130101)

Field of Classification Search

CPC: F25B (49/022); F25B (2500/222); F25B (2500/27); F25B (2600/2515)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2018/0187917	12/2017	Suzuki et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
110822760	12/2019	CN	N/A
3 480 347	12/2018	EP	N/A
3460347	12/2018	EP	N/A
3 534 087	12/2018	EP	N/A
2-140574	12/1989	JP	N/A
11-142004	12/1998	JP	N/A
2014224612	12/2013	JP	N/A
2015-209979	12/2014	JP	N/A
2019045129	12/2018	JP	N/A
WO 2013/038704	12/2012	WO	N/A
WO 2017/026147	12/2016	WO	N/A
2019116437	12/2018	WO	N/A
F/2019116437	12/2018	WO	N/A

OTHER PUBLICATIONS

Marumoto et al., Air Conditioner, Dec. 4, 2014, JP2014224612A, Whole Document (Year: 2014).
cited by examiner

Yamada et al., Air Conditioning System, Mar. 22, 2019, JP2019045129A, Whole Document (Year:
2019). cited by examiner

Extended European Search Report for European Application No. 20936720.0, dated May 23, 2023.
cited by applicant

International Preliminary Report on Patentability and English translation of the Written Opinion of
the International Searching Authority for International Application No. PCT/JP2020/019945, dated
Dec. 1, 2022. cited by applicant

European Committee for Standardization, “Refrigerating systems and heat pumps—Safety and
environmental requirements—Part 3: Installation site and personal protection,” BSI Standards
Publication, EN378-3, 2016, 34 pages total. cited by applicant

European Communication pursuant to Rule 114(2) EPC for European Application No. 20936720.0,
dated Feb. 20, 2024. cited by applicant

International Search Report, issued in PCT/JP2020/019945, PCT/ISA/210, dated Aug. 4, 2020.
cited by applicant

Written Opinion of the International Searching Authority, issued in PCT/JP2020/019945,
PCT/ISA/237, dated Aug. 4, 2020. cited by applicant

Primary Examiner: Furdge; Larry L

Attorney, Agent or Firm: BIRCH, STEWART, KOLASCH & BIRCH, LLP

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a Continuation of PCT International Application No. PCT/JP2020/019945, filed on May 20, 2020, the contents of which are hereby expressly incorporated by reference into the present application.

TECHNICAL FIELD

(1) The present disclosure relates to a refrigerant cycle apparatus.

BACKGROUND ART

(2) Patent Literature 1 discloses an air conditioner including an outdoor unit, an indoor unit, and a refrigerant circuit that circulates a refrigerant between the outdoor unit and the indoor unit. The indoor unit includes a leak sensor that detects a refrigerant leaking from the refrigerant circuit. When an abnormality such as the leak sensor reaching the end of a life has occurred, the air conditioner of Patent Literature 1 is configured to stop a compressor to stop a refrigerant flow and notify a user of the event by display or sound.

SUMMARY

(3) A refrigerant cycle apparatus of the present disclosure includes a refrigerant circuit that circulates a refrigerant, and a leak sensor that detects a refrigerant leaking from the refrigerant circuit, in which the refrigerant cycle apparatus includes, as an operating mode, a recovery mode for recognizing occurrence of an abnormality in the leak sensor and recovering a refrigerant to a predetermined location in the refrigerant circuit.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic configuration diagram illustrating a refrigerant circuit of a refrigerant cycle apparatus according to an embodiment of the present disclosure.
- (2) FIG. 2 is a block diagram of the refrigerant cycle apparatus.
- (3) FIG. 3 is a flowchart illustrating a processing procedure of a control device when a recovery mode is performed due to the end of a life of a leak sensor.
- (4) FIG. 4 is a flowchart illustrating a processing procedure of the control device when the recovery mode is performed due to failure of the leak sensor.
- (5) FIG. 5 is a table illustrating notification modes when the recovery mode is selected.
- (6) FIG. 6 is a table illustrating notification modes when a stop mode is selected.

DESCRIPTION OF EMBODIMENTS

(7) An embodiment of an air conditioning management system will be described in detail below with reference to the accompanying drawings.

(8) FIG. 1 is a schematic configuration diagram illustrating a refrigerant circuit of a refrigerant cycle apparatus according to the embodiment of the present disclosure.

- (9) As illustrated in FIG. 1, a refrigerant cycle apparatus **11** performs vapor compression refrigeration cycle operation by circulating a refrigerant through a refrigerant circuit **23**. Specifically, the refrigerant cycle apparatus **11** of the present embodiment is an air conditioner that adjusts temperature of indoor air. The air conditioner **11** includes an indoor unit (utilization-side unit) **21** and an outdoor unit (heat-source-side unit) **22**. The air conditioner **11** of the present embodiment is a multi-type air conditioner in which a plurality of indoor units **21** is connected in parallel to the outdoor unit **22**. In the present embodiment, as a refrigerant, a refrigerant having at least one of flammability, slight flammability, or toxicity, for example, an R32 refrigerant is used.
- (10) The refrigerant circuit **23** includes a compressor **30**, a four-way switching valve **32**, an outdoor heat exchanger (heat source heat exchanger) **31**, an outdoor expansion valve **34**, a liquid control valve **35**, a liquid shutoff valve **36**, indoor expansion valves **24**, indoor heat exchangers (utilization heat exchangers) **25**, a gas shutoff valve **37**, a gas control valve **38**, and refrigerant pipes **40L** and **40G** connecting these elements.
- (11) Each of the indoor units **21** includes an indoor expansion valve **24** and an indoor heat exchanger **25** included in the refrigerant circuit **23**. The indoor expansion valve **24** is formed using an electric expansion valve capable of adjusting a refrigerant pressure and a refrigerant flow rate. The indoor heat exchanger **25** is a cross fin tube type or microchannel type heat exchanger, and is used for heat exchange with indoor air.
- (12) Each of the indoor units **21** further includes an indoor fan **26** and a leak sensor **27**. The indoor fan **26** is configured to take indoor air into the indoor unit **21**, cause the indoor heat exchanger **25** to exchange heat with the taken-in air, and then blow the air into the room. The indoor fan **26** includes a motor in which the number of rotations can be adjusted by inverter control.
- (13) The leak sensor **27** detects a refrigerant leaking from the refrigerant circuit **23**. The leak sensor **27** is provided near a refrigerant pipe inside the indoor unit **21**. Alternatively, the leak sensor **27** may be provided in a remote controller **42** to be described below, a ceiling, a wall, a floor, or the like in the room.
- (14) The outdoor unit **22** includes the compressor **30**, the four-way switching valve **32**, the outdoor heat exchanger **31**, the outdoor expansion valve **34**, the liquid control valve **35**, the liquid shutoff valve **36**, the gas shutoff valve **37**, and the gas control valve **38** that are included in the refrigerant circuit **23**.
- (15) The compressor **30** sucks a low-pressure gas refrigerant and discharges a high-pressure gas refrigerant. The compressor **30** includes a motor in which the number of rotations can be adjusted by inverter control. The compressor **30** is of a variable capacity type (ability variable type) having capacity (ability) that can be changed by inverter control of the motor. Alternatively, the compressor **30** may be of a constant capacity type. Furthermore, a plurality of compressors **30** may be provided. In this case, compressors of a variable capacity type and compressors of a constant capacity type may coexist.
- (16) The four-way switching valve **32** reverses a refrigerant flow in the refrigerant pipe, and switches and supplies the refrigerant discharged from the compressor **30** to one of the outdoor heat exchanger **31** and the indoor heat exchanger **25**. As a result, the air conditioner **11** can switch between cooling operation and heating operation.
- (17) The outdoor heat exchanger **31** is, for example, a cross fin tube type or microchannel type heat exchanger, and is used for heat exchange with a refrigerant using air as a heat source.
- (18) The outdoor expansion valve **34** is formed using an electric expansion valve capable of adjusting a refrigerant pressure and a refrigerant flow rate.
- (19) The liquid shutoff valve **36** is a manual on-off valve. The gas shutoff valve **37** is also a manual on-off valve. The liquid shutoff valve **36** and the gas shutoff valve **37** are closed to block a refrigerant flow in the refrigerant pipes **40L** and **40G**, and are opened to allow a refrigerant flow in the refrigerant pipes **40L** and **40G**.
- (20) The liquid control valve **35** switches between a mode of allowing a refrigerant flow and a

mode of blocking a refrigerant flow in the liquid refrigerant pipe **40L**. The gas control valve **38** switches between a mode of allowing a refrigerant flow and a mode of blocking a refrigerant flow in the gas refrigerant pipe **40G**. In the present embodiment, as the liquid control valve **35** and the gas control valve **38**, electric expansion valves are used similarly to the outdoor expansion valve **34** and the indoor expansion valve **24**.

(21) The refrigerant pipes include the liquid refrigerant pipe **40L** and the gas refrigerant pipe **40G**. The liquid refrigerant pipe **40L** is provided between a liquid side refrigerant inlet and outlet (hereinafter, also referred to as a “liquid side end”) of the indoor heat exchanger **25** and a liquid side end of the outdoor heat exchanger **31**. The gas refrigerant pipe **40G** is provided between a gas side refrigerant inlet and outlet (hereinafter, also referred to as a “gas side end”) of the indoor heat exchanger **25** and a gas side end of the outdoor heat exchanger **31**. The liquid refrigerant pipe **40L** includes the outdoor expansion valve **34**, the gas control valve **38**, the gas shutoff valve **37**, and the indoor expansion valve **24**. The gas refrigerant pipe **40G** includes the compressor **30**, the four-way switching valve **32**, the gas control valve **38**, and the gas shutoff valve **37**. Note that, in the gas refrigerant pipe **40G**, a portion on a suction side of the compressor **30** during cooling operation to be described below is referred to as a first gas refrigerant pipe **40Ga**, and a portion on a discharge side of the compressor **30** is referred to as a second gas refrigerant pipe **40Gb**.

(22) The outdoor unit **22** further includes an outdoor fan **33**. The outdoor fan **33** includes a motor in which the number of rotations can be adjusted by inverter control. The outdoor fan **33** is configured to take outdoor air into the outdoor unit **22**, cause the outdoor heat exchanger **31** to exchange heat with the taken-in air, and then blow the air out of the outdoor unit **22**.

(23) During the cooling operation by the air conditioner **11** having the above configuration, the four-way switching valve **32** is held in a state illustrated using solid lines in FIG. **1**. A high-temperature and high-pressure gaseous refrigerant discharged from the compressor **30** flows into the outdoor heat exchanger **31** through the four-way switching valve **32**, and exchanges heat with outdoor air by operation of the outdoor fan **33** to dissipate heat. The refrigerant that has dissipated heat flows into each of the indoor units **21** through the outdoor expansion valve **34** and the liquid control valve **35** in a fully open state. In each of the indoor units **21**, the refrigerant is decompressed to a predetermined low pressure by the indoor expansion valve **24**, and the refrigerant exchanges heat with indoor air in the indoor heat exchanger **25** to evaporate. The indoor air cooled by the evaporation of the refrigerant is blown into the room by the indoor fan **26** to cool the room. The refrigerant evaporated in the indoor heat exchanger **25** returns to the outdoor unit **22** through the gas refrigerant pipe **40G**, passes through the four-way switching valve **32**, and is sucked into the compressor **30**.

(24) During heating operation by the air conditioner **11**, the four-way switching valve **32** is held in a state illustrated using broken lines in FIG. **1**. A high-temperature and high-pressure gaseous refrigerant discharged from the compressor **30** passes through the gas control valve **38** in a fully open state through the four-way switching valve **32**, and flows into the indoor heat exchanger **25** of each of the indoor units **21**. In the indoor heat exchanger **25**, the refrigerant exchanges heat with indoor air to dissipate heat. The indoor air heated by the heat dissipation of the refrigerant is blown into the room by the indoor fan **26** to heat the room. The refrigerant liquefied in the indoor heat exchanger **25** returns to the outdoor unit **22** through the liquid refrigerant pipe **40L**, is decompressed to a predetermined low pressure by the outdoor expansion valve **34**, and further exchanges heat with outdoor air in the outdoor heat exchanger **31** to evaporate. Then, the refrigerant that has evaporated and vaporized by the outdoor heat exchanger **31** is sucked into the compressor **30** through the four-way switching valve **32**.

(25) FIG. **2** is a block diagram illustrating a configuration of the refrigerant cycle apparatus. FIG. **2** illustrates two indoor units **21a** and **21b**, which may be referred to as a first indoor unit **21a** and a second indoor unit **21b**.

(26) In addition to the above configuration, each of the indoor units **21** includes an indoor control

unit **29** and the remote controller **42** (hereinafter, also simply referred to as a “remote controller”).
(27) The indoor control unit **29** is formed using a microcomputer and the like including a central processing unit (CPU) and a memory. The indoor control unit **29** controls the operation of the indoor fan **26** and the indoor expansion valve **24** described above. The indoor control unit **29** also receives a signal transmitted from the leak sensor **27**. The leak sensor **27** transmits, to the indoor control unit **29**, a detection signal indicating whether there is a refrigerant leak from the refrigerant circuit **23** and a signal indicating that the leak sensor itself has failed.

(28) The remote controller **42** is communicably connected to the indoor control unit **29**. A user can operate the remote controller **42** to turn on or off the air conditioner **11**, to input set temperature, and the like. The remote controller **42** according to the embodiment includes a display panel (display unit) **43**, a light (light emitting unit) **44**, and a buzzer (sound producing unit) **45**. As will be described below, these components form “a notification unit” that, when the leak sensor **27** has failed or reached the end of a life, notify a user or a serviceman of the event.

(29) The outdoor unit **22** includes an outdoor control unit **39** in addition to the above configuration.

(30) The outdoor control unit **39** is formed using a microcomputer and the like including a CPU and a memory. The outdoor control unit **39** controls operation of the compressor **30**, the outdoor fan **33**, the outdoor expansion valve **34**, the liquid control valve **35**, and the gas control valve **38**. The outdoor control unit **39** is communicably connected to a plurality of indoor control units **29**.

(31) [Operating Mode of Air Conditioner]

(32) The air conditioner **11** of the present embodiment includes an operating mode performed when occurrence of an abnormality of the leak sensor **27** is recognized. Specifically, the air conditioner **11** includes, as operating modes, a “stop mode” in which the operation of the air conditioner **11** is stopped when occurrence of an abnormality of the leak sensor **27** is recognized, and a “recovery mode (pump-down mode)” in which a refrigerant is recovered to a predetermined location of the refrigerant circuit **23**. The outdoor control unit **39** and the indoor control units **29** that is a control device of the air conditioner **11** perform the stop mode or the recovery mode.

(33) The stop mode and the recovery mode are selected according to user's desire, an installation environment, or the like. Specifically, the outdoor control unit **39** of the air conditioner **11** includes a selection receiving unit formed using a dip switch or the like for receiving the selection of an operating mode. A serviceman or the like can select one of the operating modes by the selection receiving unit when the air conditioner **11** is installed.

(34) The air conditioner **11** of the present embodiment recognizes that the leak sensor **27** has failed as an abnormality of the leak sensor **27**. The leak sensor **27** transmits signals illustrated in the following (a) to (c) to the indoor control unit **29**. (a) Signal indicating that the leak sensor **27** has not detected a refrigerant leak (b) Signal indicating that the leak sensor **27** has detected a refrigerant leak (c) Signal indicating failure of the leak sensor **27**

(35) When receiving the above signals of (a) and (b), the indoor control unit **29** recognizes that the leak sensor **27** functions normally. When receiving the above signal of (c), the indoor control unit **29** recognizes that the leak sensor **27** has failed. The indoor control unit **29** also recognizes that the leak sensor **27** has failed when an unknown signal that does not correspond to any of the above (a) to (c) is received and when a signal from the leak sensor **27** is interrupted.

(36) In addition to the failure as described above, the air conditioner **11** of the present embodiment also recognizes that the leak sensor **27** has reached the end of a life as an abnormality of the leak sensor **27**. The leak sensor **27** has a lifetime that is a service life set for each product. For example, since deterioration of the leak sensor **27** progresses due to energization, the lifetime is set on the basis of the integrated energization time. The indoor control unit **29** counts integrated energization time from the time point when energization to the leak sensor **27** is started. Then, when the integrated energization time exceeds a predetermined first threshold value, the indoor control unit **29** recognizes that the leak sensor **27** has reached the end of a life. Furthermore, the indoor control unit **29** recognizes that the end of a life of the leak sensor **27** is close when the integrated

energization time exceeds a predetermined second threshold value smaller than the first threshold value. The first threshold value may be, for example, 5 years, and the second threshold value may be, for example, 4 and a half years. However, the values of the first threshold value and the second threshold value are not limited, and are appropriately set according to the leak sensor 27 to be used.

(37) [Processing in Recovery Mode]

(38) (Recovery Mode Due to End of Life of Leak Sensor)

(39) Hereinafter, a processing procedure of the recovery mode by the control device 29 and 39 will be described.

(40) FIG. 3 is a flowchart illustrating a processing procedure of the control device when the recovery mode is performed due to the end of a life of a leak sensor. In FIG. 3, the two indoor units 21 included in the air conditioner 11 (the first indoor unit 21a and the second indoor unit 21b) is focused on, and an example is illustrated in which the leak sensor (first leak sensor) 27 in the one first indoor unit 21a has reached the end of a life and the leak sensor 27 (second leak sensor) 27 in the other second indoor unit 21b has not reached the end of a life.

(41) The indoor control units 29 of the first indoor unit 21a and the second indoor unit 21b count integrated energization time T.sub.1 and T.sub.2 of the leak sensors 27, respectively (steps S21 and S31). When the integrated energization time T.sub.1 of the leak sensor 27 in the first indoor unit 21a exceeds a second threshold value T.sub.th2, the indoor control unit 29 transmits the information to the outdoor control unit 39 of the outdoor unit 22 (step S22). The outdoor control unit 39 transmits an instruction to perform a lifetime preliminary notification to the indoor control unit 29 of the first indoor unit 21a on the basis of the information received from the indoor control unit 29 (step S11).

(42) On the basis of the instruction from the outdoor control unit 39, the indoor control unit 29 of the first indoor unit 21a notifies that the end of a life of the leak sensor 27 is close using at least one of the display panel 43, the light 44, or the buzzer 45 in the remote controller 42 (step S23). A user, a serviceman, and the like can grasp that the end of a life of the leak sensor 27 of the first indoor unit 21a is close on the basis of the notification by the remote controller 42, and can take measures such as preparation for replacing the leak sensor 27 in advance. A specific mode of the notification will be described below.

(43) Next, when the integrated energization time T.sub.1 of the leak sensor 27 in the first indoor unit 21a exceeds a first threshold value T.sub.th1, the indoor control unit 29 transmits the information to the outdoor control unit 39 (step S24). The outdoor control unit 39 transmits an instruction to perform a lifetime notification to each of the indoor control units 29 of the first indoor unit 21a and the second indoor unit 21b on the basis of the information received from the indoor control unit 29 (step S12).

(44) Each of the indoor control units 29 of the first indoor unit 21a and the second indoor unit 21b that have received the instruction from the outdoor control unit 39 notifies that the leak sensor 27 of the first indoor unit 21a has reached the end of a life using at least one of the display panel 43, the light 44, or the buzzer 45 in the remote controller 42 (steps S25 and S32).

(45) Note that the indoor control unit 29 of the first indoor unit 21a may perform the notification not by the instruction from the outdoor control unit 39 (steps S11 and S12) but by a fact that the integrated energization time T.sub.1 of the leak sensor 27 exceeds the second threshold value T.sub.th2 or the first threshold value T.sub.th1.

(46) Next, upon recognizing that the leak sensor 27 of the first indoor unit 21a has reached the end of a life, the outdoor control unit 39 performs the recovery mode. Specifically, when the air conditioner 11 is in the cooling operation or the heating operation, the outdoor control unit 39 temporarily stops the compressor 30 (step S14). Then, the four-way switching valve 32 is switched to a mode similar to the cooling operation, the liquid control valve 35 is closed, and then the compressor 30 is operated (steps S15 and S16). The outdoor control unit 39 transmits an instruction to operate the indoor fan 26 to each of the indoor control units 29, and each of the indoor control

units **29** operates the indoor fan **26** when the indoor fan **26** is stopped and continues the operation as it is when the indoor fan **26** is operating (steps **S26** and **S33**).

(47) According to the above operation, a refrigerant in the refrigerant circuit **23** is recovered to the outdoor heat exchanger **31** side between the liquid control valve **35** and the gas control valve **38**. The refrigerant is mainly recovered inside the outdoor heat exchanger **31**. When the refrigerant is recovered to the outdoor heat exchanger **31** side, the outdoor control unit **39** closes the gas control valve **38** and stops the compressor **30** (step **S17**), and transmits a notification of the end of the recovery mode to the indoor control units **29** (step **S18**). Each of indoor heat exchangers **25** stops the indoor fan **26** on the basis of the notification of the end of the recovery mode. Thus, all processing related to the recovery mode ends.

(48) As described above, since the air conditioner **11** according to the present embodiment recovers the refrigerant as the leak sensor **27** has reached the end of a life, leakage of the refrigerant can be prevented even if the refrigerant pipes **40L** and **40G** are damaged by subsequent replacement work of the leak sensor **27** or other maintenance work.

(49) When the leak sensor **27** reaches the end of a life, the refrigerant may not be accurately detected even if the refrigerant leaks, and thus, during performance of the recovery mode, the indoor fan **26** is operated in each of the indoor units **21** to diffuse the leaking refrigerant, thereby preventing the refrigerant from staying in the room. However, the indoor fan **26** may not necessarily be operated during the performance of the recovery mode.

(50) In the recovery mode, the refrigerant in the refrigerant circuit **23** may be recovered to an accumulator or a receiver (neither is illustrated) included in the refrigerant circuit **23** instead of or in addition to the outdoor heat exchanger **31**.

(51) Although the compressor **30** is temporarily stopped when the recovery mode is started in the above processing procedure, when the compressor **30** is operated by the cooling operation or the heating operation, the operating mode may be shifted to the recovery mode without the compressor **30** being stopped. In this case, the normal operation can be quickly shifted to the operation in the recovery mode.

(52) Although a case where the leak sensor **27** of the first indoor unit **21a** has reached the end of a life has been described in the above description, similar processing is performed when the leak sensor (second leak sensor) **27** of the second indoor unit **21b** has reached the end of a life.

(53) In a case where the air conditioner **11** includes three or more indoor units **21**, the above processing of the first indoor unit **21a** is performed in an indoor unit **21** in which the leak sensor **27** has reached the end of a life, and the above processing of the second indoor unit **21b** is performed in indoor units **21** in which the leak sensors **27** have not reached the end of a life.

(54) (Recovery Mode Due to Failure of Leak Sensor)

(55) FIG. **4** is a flowchart illustrating a processing procedure of the control device when the recovery mode is performed due to failure of a leak sensor.

(56) As described above, the indoor control units **29** of the first indoor unit **21a** and the second indoor unit **21b** receive signals indicating whether there is a refrigerant leak from the leak sensors **27**. Upon recognizing failure of the leak sensor **27**, the indoor control unit **29** in the first indoor unit **21a** transmits the information to the outdoor control unit **39** (step **S51**).

(57) The outdoor control unit **39** transmits an instruction to perform a failure notification to the indoor control units **29** of the first indoor unit **21a** and the second indoor unit **21b** on the basis of the information received from the indoor control unit **29** (step **S41**). On the basis of the instruction of the outdoor control unit **39**, each of the indoor control units **29** of the first indoor unit **21a** and the second indoor unit **21b** notifies that the leak sensor **27** of the first indoor unit **21a** has failed using at least one of the display panel **43**, the light **44**, or the buzzer **45** in the remote controller **42** (steps **S53** and **S62**). A user, a serviceman, and the like can grasp that the leak sensor **27** has failed on the basis of the notification in the remote controller **42**.

(58) Next, upon recognizing that the leak sensor **27** has failed, the outdoor control unit **39** performs

the recovery mode. The processing of the outdoor control unit **39** (S42 to S47) and the processing of the indoor control units **29** (S53 to S54 and S62 to S63) in the recovery mode are the same as the processing in the recovery mode due to the end of a life of the leak sensor **27**.

(59) [Processing in Stop Mode]

(60) As described above, the air conditioner **11** of the present embodiment includes the recovery mode and the stop mode as the operating modes when an abnormality occurs in a leak sensor **27**, and can select one of the two modes. When the stop mode is selected, the stop mode is performed instead of the recovery mode (steps S13 and S42) in FIGS. 3 and 4, and the control device **29** and **39** stop the compressor **30**, the outdoor fan **33**, the indoor fans **26**, and the like and close the liquid control valve **35** and the gas control valve **38**.

(61) [Modes of Abnormality Notification]

(62) FIG. 5 is a table illustrating notification modes when the recovery mode is selected. As described above, in the remote controllers **42** of the first indoor unit **21a** and the second indoor unit **21b**, when a leak sensor **27** has failed, reached the end of a life, or come close to the end of a life, a notification indicating these abnormalities is performed. FIG. 5 illustrates specific modes of the notification according to respective cases.

(63) When the leak sensor **27** of the first indoor unit **21a** has failed and the leak sensor **27** of the second indoor unit **21b** has not failed, abnormality display A indicating the failure of the leak sensor **27** is displayed on the display panel **43** of the first indoor unit **21a**, and at the same time, the light **44** emits light and the buzzer **45** sounds. At this time, abnormality display C indicating the abnormality of the leak sensor **27** of the other indoor unit (first indoor unit) **21a** is displayed on the display panel **43** of the second indoor unit **21b**, and the light **44** and the buzzer **45** do not operate.

(64) As the first indoor unit **21a** and the second indoor unit **21b** perform different notifications in this manner, a user and a serviceman can recognize that a failure has occurred in the leak sensor **27** in the first indoor unit **21a** by the notification from the first indoor unit **21a**, and can recognize that an abnormality has occurred in an indoor unit other than the second indoor unit **21b** connected to the second indoor unit **21b** by the notification from the second indoor unit **21b**. In a case where there are two indoor units, a user and a serviceman can recognize that an abnormality has occurred in the first indoor unit **21a** by not only the notification from the first indoor unit **21a** but also the notification from the second indoor unit **21b**.

(65) When the leak sensor **27** of the first indoor unit **21a** has reached the end of a life and the leak sensor **27** of the second indoor unit **21b** has not reached the end of a life, abnormality display B indicating that the leak sensor **27** has reached the end of a life is displayed on the display panel **43** of the first indoor unit **21a**, and at the same time, the light **44** emits light and the buzzer **45** sounds. The abnormality display C indicating the abnormality of the leak sensor **27** of the other indoor unit (first indoor unit) **21a** is displayed on the display panel **43** of the second indoor unit **21b**, and the light **44** and the buzzer **45** do not operate.

(66) As the first indoor unit **21a** and the second indoor unit **21b** perform different notifications in this manner, a user and a serviceman can recognize that the leak sensor **27** has reached the end of a life in the first indoor unit **21a** by the notification from the first indoor unit **21a**, and can recognize that an abnormality has occurred in an indoor unit other than the second indoor unit **21b** connected to the second indoor unit **21b** by the notification from the second indoor unit **21b**. In a case where there are two indoor units, a user and a serviceman can recognize that an abnormality has occurred in the first indoor unit **21a** by not only the notification from the first indoor unit **21a** but also the notification from the second indoor unit **21b**.

(67) As described above, when an abnormality occurs in the leak sensor **27**, the recovery mode is performed and the cooling operation and the heating operation by the air conditioner **11** are disabled, and the operation of the air conditioner **11** is substantially prohibited. Therefore, a user and the like can grasp that the operation of the air conditioner **11** is prohibited not only by the notification by the first indoor unit **21a** but also by the notification by the second indoor unit **21b**.

(68) When the end of a life of the leak sensor 27 of the first indoor unit 21a is close and the end of a life of the leak sensor 27 of the second indoor unit 21b is not close, maintenance display indicating that the end of a life of the leak sensor 27 is close is displayed on the display panel 43 of the first indoor unit 21a, and the light 44 and the buzzer 45 do not operate. The display panel 43 of the second indoor unit 21b does not display the abnormality, and the light 44 and the buzzer 45 do not operate.

(69) As described above, the notification is performed only by the first indoor unit 21a including the leak sensor 27 that has come close to the end of a life, so that a user and a serviceman can easily grasp that the end of a life of the leak sensor 27 of the first indoor unit 21a is close.

(70) FIG. 6 is a table illustrating notification modes when the stop mode is selected.

(71) When the leak sensor 27 of the first indoor unit 21a has failed and the leak sensor 27 of the second indoor unit 21b has not failed, the abnormality display A indicating the failure of the leak sensor 27 is displayed on the display panel 43 of the first indoor unit 21a, and at the same time, the light 44 emits light and the buzzer 45 sounds. At this time, the display panel 43 of the second indoor unit 21b does not display an indication of the abnormality, and the light 44 and the buzzer 45 do not operate. However, when a user or the like operates the remote controller 42, an error is displayed on the display panel 43. A user and a serviceman can recognize that the leak sensor 27 of the first indoor unit 21a has failed by the notification from the first indoor unit 21a. Furthermore, a user and a serviceman can recognize that an abnormality has occurred in the leak sensor 27 of an indoor unit other than the second indoor unit 21b connected to the second indoor unit 21b by the notification (error display) of the second indoor unit 21b when operating the remote controller 42. In a case where there are two indoor units, a user and a serviceman can recognize that an abnormality has occurred in the first indoor unit 21a by not only the notification from the first indoor unit 21a but also the notification from the second indoor unit 21b.

(72) When the leak sensor 27 of the first indoor unit 21a has reached the end of a life and the leak sensor 27 of the second indoor unit 21b has not reached the end of a life, the abnormality display B indicating that the leak sensor 27 has reached the end of a life is displayed on the display panel 43 of the first indoor unit 21a, and at the same time, the light 44 emits light and the buzzer 45 sounds. At this time, the display panel 43 of the second indoor unit 21b does not display an indication of the abnormality, and the light 44 and the buzzer 45 do not operate. However, when a user or the like operates the remote controller 42, an error is displayed. A user and a serviceman can recognize that the leak sensor 27 of the first indoor unit 21a has reached the end of a life by the notification from the first indoor unit 21a. Furthermore, a user and a serviceman can recognize that an abnormality has occurred in the leak sensor 27 of an indoor unit other than the second indoor unit 21b connected to the second indoor unit 21b by the notification (error display) of the second indoor unit 21b when operating the remote controller 42. In a case where there are two indoor units, a user and a serviceman can recognize that an abnormality has occurred in the first indoor unit 21a by not only the notification from the first indoor unit 21a but also the notification from the second indoor unit 21b.

(73) As described above, in the stop mode, when an abnormality occurs in the leak sensor 27 of the first indoor unit 21a, an error is displayed on the remote controller 42 of the second indoor unit 21b only when a user or the like operates the remote controller 42. As a result, only a user who attempts to operate the air conditioner 11 is notified of occurrence of an abnormality, and other user is not notified of the occurrence of the abnormality. Therefore, unnecessary notification is not performed to the other user, and the possibility of giving a sense of anxiety to the other user can be reduced.

(74) When the end of a life of the leak sensor 27 of the first indoor unit 21a is close and the end of a life of the leak sensor 27 of the second indoor unit 21b is not close, the maintenance display indicating that the end of a life of the leak sensor 27 is close is displayed on the display panel 43 of the first indoor unit 21a, and the light 44 and the buzzer 45 do not operate. The display panel 43 of the second indoor unit 21b does not display the abnormality, and the light 44 and the buzzer 45 do

not operate.

(75) As described above, the notification is performed only by the first indoor unit **21a** including the leak sensor **27** that has come close to the end of a life, so that a user and a serviceman can easily grasp that the end of a life of the leak sensor **27** of the first indoor unit **21a** is close.

(76) The abnormality display A, B, and C displayed on the display panel **43** of the remote controller **42** may be codes indicating an abnormality or contents of the abnormality. The maintenance display and the error display displayed on the display panel **43** may be display of codes or display of contents themselves.

(77) The notification modes of the notification unit **43**, **44**, and **45** when an abnormality occurs in the leak sensor **27** can be variously changed. For example, the first indoor unit **21a** in which an abnormality has occurred in the leak sensor **27** and the second indoor unit **21b** in which no abnormality has occurred may perform notifications by different methods using the same means among the display panel **43**, the light **44**, and the buzzer **45**. For example, when both the first indoor unit **21a** and the second indoor unit **21b** perform notifications using lights **44**, one light **44** can be blinked and the other light **44** can be turned on. Furthermore, the first indoor unit **21a** in which an abnormality has occurred in the leak sensor **27** and the second indoor unit **21b** in which no abnormality has occurred may perform notifications using different means among the display panel **43**, the light **44**, and the buzzer **45**. For example, the light **44** may be turned on in the first indoor unit **21a**, and the buzzer **45** may be operated in the second indoor unit **21b**.

Effects of Embodiments

(78) (1) A refrigerant cycle apparatus **11** of the above embodiment includes a refrigerant circuit **23** that circulates a refrigerant, and a leak sensor **27** that detects a refrigerant leaking from the refrigerant circuit **23**, in which the refrigerant cycle apparatus **11** includes, as an operating mode, a recovery mode for recognizing occurrence of an abnormality in the leak sensor **27** and recovering a refrigerant to a predetermined location in the refrigerant circuit **23**. As a result, leakage of the refrigerant from the refrigerant circuit **23** due to replacement of the leak sensor **27**, other maintenance, or the like can be prevented.

(79) (2) The refrigerant cycle apparatus **11** of the above embodiment further includes a control device (indoor control unit **29** and outdoor control unit **39**) that control the refrigerant circuit **23**. The control device **29** and **39** perform the recovery mode. The refrigerant cycle apparatus **11** further includes, as an operating mode, a stop mode for recognizing occurrence of an abnormality in the leak sensor **27** and stopping a refrigerant flow in the refrigerant circuit **23**, in which the control device **29** and **39** perform the stop mode. The control device **29** and **39** receive selection of an operating mode to be performed when occurrence of an abnormality of the leak sensor **27** is recognized from the recovery mode and the stop mode. As a result, one of the recovery mode and the stop mode can be selected to be performed when an abnormality has occurred in the leak sensor **27** according to the installation environment of the refrigerant cycle apparatus **11** or the like.

(80) (3) The refrigerant circuit **23** of the above embodiment includes a heat source heat exchanger **31**, a utilization heat exchanger **25**, a compressor **30**, a liquid refrigerant pipe **40L** connecting a liquid side end of the utilization heat exchanger **25** and a liquid side end of the heat source heat exchanger **31**, a first gas refrigerant pipe **40Ga** connecting a gas side end of the utilization heat exchanger **25** and a suction pipe of the compressor **30**, a second gas refrigerant pipe **40Gb** connecting a gas side end of the heat source heat exchanger **31** and a discharge pipe of the compressor **30**, a liquid control valve **35** included in the liquid refrigerant pipe **40L**, and a gas control valve **38** included in the first gas refrigerant pipe **40Ga**, and the control device **29** and **39** close the liquid control valve **35** and operate the compressor **30** at a start of operation in the recovery mode, and close the gas control valve **38** at an end of operation in the recovery mode. As described above, a refrigerant can be recovered to the heat source heat exchanger **31** side between the liquid control valve **35** and the gas control valve **38** because the control device **29** and **39** close the liquid control valve **35** when the operation in the recovery mode is started, and the refrigerant

can be prevented from flowing back to the utilization heat exchanger 25 side because the gas control valve 38 is closed when the operation in the recovery mode is ended.

(81) (4) In the above embodiment, the abnormality of the leak sensor 27 is failure of the leak sensor 27 or an end of a life of the leak sensor 27. The refrigerant cycle apparatus 11 further includes a notification unit (display panel 43, light 44, and buzzer 45), in which the control device 29 and 39 recognize an end of a life of the leak sensor 27 when integrated energization time of the leak sensor 27 exceeds a predetermined first threshold value T.sub.th1, and the notification unit 43, 44, and 45 perform different notifications between a case where the integrated energization time exceeds the first threshold value T.sub.th1 and a case where the integrated energization time exceeds a predetermined second threshold value T.sub.th2 shorter than the first threshold value T.sub.th1. As a result, a serviceman and a user can grasp not only that the leak sensor 27 has reached the end of a life but also that the end of a life of the leak sensor 27 is close by the notification from the notification unit 43, 44, and 45.

(82) (5) In the above embodiment, the refrigerant cycle apparatus 11 includes an outdoor unit 22 in which a refrigerant is circulated by the refrigerant circuit 23, and a first indoor unit 21a and a second indoor unit 21b in which a refrigerant is circulated by the refrigerant circuit 23. The first indoor unit 21a includes a leak sensor (first leak sensor) 27 and a notification unit (first notification unit) 43, 44, and 45, and the second indoor unit 21b includes a leak sensor (second leak sensor) 27 and a notification unit (second notification unit) 43, 44, and 45. When an abnormality has occurred in the leak sensor 27 in the first indoor unit 21a and no abnormality has occurred in the leak sensor 27 in the second indoor unit 21b, the notification unit 43, 44, and 45 of the first indoor unit 21a and the notification unit 43, 44, and 45 of the second indoor unit 21b perform different notifications. As described above, a serviceman and a user can grasp that the abnormality has occurred in the leak sensor 27 of the first indoor unit 21a not only by the notification unit 43, 44, and 45 of the first indoor unit 21a but also by the notification unit 43, 44, and 45 of the second indoor unit 21b because the notification unit 43, 44, and 45 of the first indoor unit 21a including the leak sensor 27 in which the abnormality has occurred and the notification unit 43, 44, and 45 of the second indoor unit 21b perform different notifications.

(83) (6) In the above embodiment, the notification unit 43, 44, and 45 of the first indoor unit 21a and the notification unit 43, 44, and 45 of the second indoor unit 21b perform notifications by different means among a plurality of means including light, sound, and display, notifications by different combinations of the plurality of means, or notifications by different notification methods of a same means. As a result, when an abnormality has occurred in the leak sensor 27 of the first indoor unit 21a, a serviceman and a user can clearly identify the difference between a notification by the notification unit 43, 44, and 45 of the first indoor unit 21a and a notification by the notification unit 43, 44, and 45 of the second indoor unit 21b.

(84) (7) In the above embodiment, the control device 29 and 39 perform the recovery mode while continuing the refrigerant flow in a case of recognizing occurrence of an abnormality in the leak sensor 27 while a refrigerant is circulating in the refrigerant circuit 23. In this case, the normal operation can be quickly shifted to the recovery mode when an abnormality has occurred in the leak sensor 27.

(85) The present disclosure should not be limited to the above exemplification, but is intended to include any modification recited in claims within meanings and a scope equivalent to those of the claims.

REFERENCE SIGNS LIST

(86) 11 Refrigerant cycle apparatus (air conditioner) 21 Indoor unit 22 Outdoor unit 23 Refrigerant circuit 25 Indoor heat exchanger (utilization heat exchanger) 27 Leak sensor 29 Indoor control unit (control device) 30 Compressor 31 Outdoor heat exchanger (heat source heat exchanger) 35 Liquid control valve 38 Gas control valve 39 Outdoor control unit (control device) 40Ga First gas refrigerant pipe 40Gb Second gas refrigerant pipe 40L Liquid refrigerant pipe 43 Display panel

(notification unit) **44** Light (notification unit) **45** Buzzer (notification unit) T.sub.1 Integrated energization time T.sub.2 Integrated energization time T.sub.th1 First threshold value T.sub.th2 Second threshold value

CITATION LIST

Patent Literature

(87) Patent Literature 1: WO 2017/026147 A

Claims

1. A refrigerant cycle apparatus comprising: a refrigerant circuit that circulates a refrigerant; and a leak sensor that detects a refrigerant leaking from the refrigerant circuit, wherein the refrigerant cycle apparatus includes an operating mode that is performed when recognizing an occurrence of an abnormality in the leak sensor, and the operating mode includes a recovery mode for recovering a refrigerant to a predetermined location in the refrigerant circuit.
2. The refrigerant cycle apparatus according to claim 1 further comprising a controller that controls the refrigerant circuit, wherein the controller performs the recovery mode.
3. The refrigerant cycle apparatus according to claim 2 further comprising, as an operating mode, a stop mode for recognizing occurrence of an abnormality in the leak sensor and stopping a refrigerant flow in the refrigerant circuit, wherein the controller performs the stop mode.
4. The refrigerant cycle apparatus according to claim 3, wherein the controller receives selection of an operating mode to be performed when occurrence of an abnormality of the leak sensor is recognized from the recovery mode and the stop mode.
5. The refrigerant cycle apparatus according to claim 2, wherein the refrigerant circuit comprises a heat source heat exchanger, a utilization heat exchanger, a compressor, a liquid refrigerant pipe connecting a liquid side end of the utilization heat exchanger and a liquid side end of the heat source heat exchanger, a first gas refrigerant pipe connecting a gas side end of the utilization heat exchanger and a suction pipe of the compressor, a second gas refrigerant pipe connecting a gas side end of the heat source heat exchanger and a discharge pipe of the compressor, a liquid control valve included in the liquid refrigerant pipe, and a gas control valve included in the first gas refrigerant pipe, and the controller closes the liquid control valve and operates the compressor at a start of operation in the recovery mode, and closes the gas control valve at an end of operation in the recovery mode.
6. The refrigerant cycle apparatus according to claim 1, wherein the abnormality is failure of the leak sensor or an end of a life of the leak sensor.
7. The refrigerant cycle apparatus according to claim 6 further comprising a notification unit, wherein the refrigerant cycle apparatus recognizes an end of a life of the leak sensor when integrated energization time of the leak sensor exceeds a predetermined first threshold value, and the notification unit performs different notifications between a case where the integrated energization time exceeds the first threshold value and a case where the integrated energization time exceeds a predetermined second threshold value shorter than the first threshold value.
8. The refrigerant cycle apparatus according to claim 1 further comprising: an outdoor unit in which a refrigerant is circulated by the refrigerant circuit; and a first indoor unit and a second indoor unit in which a refrigerant is circulated by the refrigerant circuit, wherein the first indoor unit includes a first leak sensor and a first notification unit, and the second indoor unit includes a second leak sensor and a second notification unit, and wherein, when an abnormality has occurred in the first leak sensor and no abnormality has occurred in the second leak sensor, the first notification unit and the second notification unit perform different notifications.
9. The refrigerant cycle apparatus according to claim 8, wherein the first notification unit and the second notification unit perform notifications by different means among a plurality of means including light, sound, and display, notifications by different combinations of the plurality of

means, or notifications by different notification methods of a same means.

10. The refrigerant cycle apparatus according to claim 2, wherein the controller performs the recovery mode while continuing the refrigerant flow in a case of recognizing occurrence of an abnormality in the leak sensor while a refrigerant is circulating in the refrigerant circuit.

11. The refrigerant cycle apparatus according to claim 3, wherein the refrigerant circuit comprises a heat source heat exchanger, a utilization heat exchanger, a compressor, a liquid refrigerant pipe connecting a liquid side end of the utilization heat exchanger and a liquid side end of the heat source heat exchanger, a first gas refrigerant pipe connecting a gas side end of the utilization heat exchanger and a suction pipe of the compressor, a second gas refrigerant pipe connecting a gas side end of the heat source heat exchanger and a discharge pipe of the compressor, a liquid control valve included in the liquid refrigerant pipe, and a gas control valve included in the first gas refrigerant pipe, and the controller closes the liquid control valve and operates the compressor at a start of operation in the recovery mode, and closes the gas control valve at an end of operation in the recovery mode.

12. The refrigerant cycle apparatus according to claim 4, wherein the refrigerant circuit comprises a heat source heat exchanger, a utilization heat exchanger, a compressor, a liquid refrigerant pipe connecting a liquid side end of the utilization heat exchanger and a liquid side end of the heat source heat exchanger, a first gas refrigerant pipe connecting a gas side end of the utilization heat exchanger and a suction pipe of the compressor, a second gas refrigerant pipe connecting a gas side end of the heat source heat exchanger and a discharge pipe of the compressor, a liquid control valve included in the liquid refrigerant pipe, and a gas control valve included in the first gas refrigerant pipe, and the controller closes the liquid control valve and operates the compressor at a start of operation in the recovery mode, and closes the gas control valve at an end of operation in the recovery mode.

13. The refrigerant cycle apparatus according to claim 2, wherein the abnormality is failure of the leak sensor or an end of a life of the leak sensor.

14. The refrigerant cycle apparatus according to claim 3, wherein the abnormality is failure of the leak sensor or an end of a life of the leak sensor.

15. The refrigerant cycle apparatus according to claim 4, wherein the abnormality is failure of the leak sensor or an end of a life of the leak sensor.

16. The refrigerant cycle apparatus according to claim 5, wherein the abnormality is failure of the leak sensor or an end of a life of the leak sensor.

17. The refrigerant cycle apparatus according to claim 2 further comprising: an outdoor unit in which a refrigerant is circulated by the refrigerant circuit; and a first indoor unit and a second indoor unit in which a refrigerant is circulated by the refrigerant circuit, wherein the first indoor unit includes a first leak sensor and a first notification unit, and the second indoor unit includes a second leak sensor and a second notification unit, and wherein, when an abnormality has occurred in the first leak sensor and no abnormality has occurred in the second leak sensor, the first notification unit and the second notification unit perform different notifications.

18. The refrigerant cycle apparatus according to claim 3 further comprising: an outdoor unit in which a refrigerant is circulated by the refrigerant circuit; and a first indoor unit and a second indoor unit in which a refrigerant is circulated by the refrigerant circuit, wherein the first indoor unit includes a first leak sensor and a first notification unit, and the second indoor unit includes a second leak sensor and a second notification unit, and wherein, when an abnormality has occurred in the first leak sensor and no abnormality has occurred in the second leak sensor, the first notification unit and the second notification unit perform different notifications.

19. The refrigerant cycle apparatus according to claim 4 further comprising: an outdoor unit in which a refrigerant is circulated by the refrigerant circuit; and a first indoor unit and a second indoor unit in which a refrigerant is circulated by the refrigerant circuit, wherein the first indoor unit includes a first leak sensor and a first notification unit, and the second indoor unit includes a

second leak sensor and a second notification unit, and wherein, when an abnormality has occurred in the first leak sensor and no abnormality has occurred in the second leak sensor, the first notification unit and the second notification unit perform different notifications.

20. The refrigerant cycle apparatus according to claim 5 further comprising: an outdoor unit in which a refrigerant is circulated by the refrigerant circuit; and a first indoor unit and a second indoor unit in which a refrigerant is circulated by the refrigerant circuit, wherein the first indoor unit includes a first leak sensor and a first notification unit, and the second indoor unit includes a second leak sensor and a second notification unit, and wherein, when an abnormality has occurred in the first leak sensor and no abnormality has occurred in the second leak sensor, the first notification unit and the second notification unit perform different notifications.
